

Retrieval-Augmented Generation (RAG) in Financial & Economic Analysis

Module 3: Grounded Knowledge Extraction, GraphRAG, and LightRAG

Prof. Chae Shick Chung

Graduate School of Economics / MBA
Sogang University

Fall 2026

Executive Summary & Learning Objectives

Core Objective

Master **Retrieval-Augmented Generation (RAG)** as a precision instrument to eliminate LLM hallucinations, enforce 100% source traceability, and automate complex financial research.

Key Concepts Covered:

- LLM Hallucinations & In-Context Grounding
- Architecture of **Naive RAG** (4-Step Pipeline)
- Structural Paradigms: **GraphRAG** & **LightRAG**
- Side-by-Side Architectural Comparison

Financial Applications:

- 10-K Filings & Notes Parsing (Covenants/R&D)
- Cross-Examination of Macro Reports
- Compliance & Regulatory Checking
- **Human-in-the-Loop (HITL)** 3-Step Editorship

The Imperative for RAG in Financial Analysis

The High Cost of Hallucinations in Finance

Standard LLMs rely on static parametric memory. In finance, a **1% numerical error** or fabricated clause can cause millions in investment losses or severe regulatory penalties.

Parametric LLM Limitations

- Knowledge cutoff & model drift
- Hallucination under complex queries
- Unverifiable "Black-Box" outputs
- Exposure of confidential data via APIs

RAG Solution Architecture

- **External Knowledge Base:** Ingests 10-K, SEC filings, news
- **In-Context Grounding:** Restricts LLM to provided text
- **Source Traceability:** Hyperlinked citations [Doc, p.N]
- **Private Deployment:** On-premise SLMs

Architecture of Naive RAG (Chunk-Based)

The 4-Step Sequential Pipeline

1. Query & 2. Retrieval

- **Query:** Analyst inputs targeted question (e.g., "CAPEX changes in Q2").
- **Chunking & Indexing:** Documents split into fixed-length text chunks (500 tokens).
- **Semantic Search:** Vector DB retrieves Top- K relevant chunks via cosine similarity.

3. Grounding & 4. Generation

- **In-Context Grounding:** Injected retrieved passages into system prompt.
- **Prompt Constraint:** "Answer strictly using provided context; do not extrapolate."
- **Generation:** LLM synthesizes precise, cited financial summary.

Vulnerabilities of Naive RAG in Financial Workflows

Why Fixed Chunks Fail Complex Financial Analysis

Naive RAG treats documents as isolated text snippets. It struggles with multi-hop reasoning, table structures, and document-wide relationships.

- **Retrieval Omission:** Numerical tables in 10-K filings often lack semantic context, causing vector search to miss critical balance sheet items.
- **Stale Data / Model Drift:** When multiple quarterly reports exist, chunk search may retrieve outdated figures from 2025 instead of 2026.
- **Context Mixing (Attribution Fallacy):** The LLM combines facts from Firm A's report and Firm B's report into a single erroneous causal statement.
- **Loss of Global Perspective:** Cannot answer high-level questions like "Summarize key risk factors across the entire energy portfolio."

GraphRAG (Microsoft): Knowledge Graphs & Global Queries

Structural Graph Paradigm

LLMs extract **Entities** (Companies, Executives, Products) and **Relations** (Mergers, Supply Chain, Litigation) to build a **Knowledge Graph (KG)**.

Key Innovations:

- **Community Clustering:** Applies Louvain/Leiden algorithms to group related nodes.
- **Pre-computed Summaries:** Generates hierarchical summaries for each community.
- **Global Queries:** Excels at macro, document-wide synthesis questions.

Financial Trade-offs:

- **Extreme Indexing Cost:** Requires massive LLM API calls to build graphs.
- **Batch-Only Nature:** Cannot handle real-time streaming market data.
- **Re-indexing Overhead:** Adding a single news item requires costly graph regeneration.

LightRAG: Dual-Level Retrieval & Incremental Updates

Lightweight Dual-Level Graph Architecture

Designed to solve GraphRAG's high indexing cost and latency issues while preserving relational intelligence for fast-moving financial markets.

Dual-Level Retrieval

- **Low-Level Retrieval:** Targets specific entities, exact financial metrics, and local relations.
- **High-Level Retrieval:** Captures overarching themes, macro sentiment, and broad trends.
- **Combined Context:** Integrates both levels into a unified generation prompt.

Incremental Graph Updates

- **Dynamic Ingestion:** Seamlessly merges new incoming news/tweets into existing Key-Value KV store.
- **Zero Full Re-indexing:** Avoids rebuilding the entire graph from scratch.
- **Trading Desk Fit:** Ideal for intra-day news feeds and risk dashboards.

Architectural Matrix: Naive RAG vs. GraphRAG vs. LightRAG

Feature	Naive RAG	GraphRAG (MS)	LightRAG
Data Structure	Text Chunks	Knowledge Graph + Summaries	Dual-Level KV Knowledge Graph
Indexing Cost	Very Low	Extremely High (Heavy LLM)	Moderate / Low
Incremental Updates	Immediate (Append chunk)	Not Supported (Batch rebuild)	Supported (Seamless KV merge)
Global Queries	Poor (Fragmented context)	Excellent (Community-level)	Good (High-level retrieval)
Local Queries	Good (Exact keyword)	Good (Entity-focused)	Excellent (Low-level retrieval)
Financial Best-Fit	Static manuals, policy SOPs	Annual 10-K macro synthesis	Real-time news & trading desk

Real-World Financial Use Cases

1. Corporate Filings (10-K)

- Automated tracking of debt covenants.
- Uncovering hidden footnotes in R&D and inventory accounting.
- Parsing liquidity risk clauses across quarterly reports.

2. Macro Cross-Exam

- Cross-referencing Fed FOMC minutes, IMF Outlook, and BOK reports.
- Identifying conflicting interest rate projections across top IBs.
- Stress-testing tail risk scenarios.

3. Compliance Check

- Ingesting Financial Consumer Protection Act laws.
- Auditing new financial product proposals against regulatory SOPs.
- Pre-filtering legal litigation risk prior to human sign-off.

Human-in-the-Loop (HITL) & 3-Step Editorship Protocol

Fiduciary Responsibility in AI-Assisted Research

AI generates draft insights, but human financial analysts bear 100% legal, ethical, and fiduciary liability for investment recommendations.

Step 1: Traceability Audit

Cross-check every decimal, percentage, and ratio against the primary source PDF via hyperlinked citations.

Step 2: Caveat Restorations

Detect if the LLM over-summarized and omitted crucial author caveats, scope limits, or interest rate conditions.

Step 3: Fiduciary Sign-Off

Enforce formal human reviewer endorsement prior to publishing research to clients or trading desks.

Seminar Discussion Topics & Case Studies

Discussion 1: The Accounting Footnote Trap

A company's 10-K summary shows a 25% spike in R&D investment, hailed by AI as an innovation driver. Deep footnote inspection reveals \$40M of this "R&D" was an one-off patent litigation settlement, while actual lab CAPEX fell by 12%.

- How can hybrid retrieval (BM25 + Semantic) and prompt guardrails prevent LLM gullibility?

Discussion 2: RAG Architecture Selection for Institutional Desks

As Chief Digital Officer (CDO) of an asset management firm, how do you balance Capex, Opex, and real-time latency when choosing between Naive RAG, GraphRAG, LightRAG, and Multi-Agent frameworks?

Summary & Transition to Module 4

Module 3 Takeaways

- **Grounding is Non-Negotiable:** RAG is the essential bridge connecting parametric LLMs with verifiable primary financial sources.
- **Match Architecture to Problem:** Naive RAG for static SOPs; GraphRAG for macro 10-K synthesis; LightRAG for real-time market feeds.
- **Analyst as Chief Editor:** Shift from manual document parser to critical reviewer enforcing 3-step editorship.

Next Horizon: Module 4 – Autonomous AI Agents (Hermes, Buzz, GrokBot)

From passive retrieval to autonomous goal decomposition, tool execution, and behavioral market simulation.